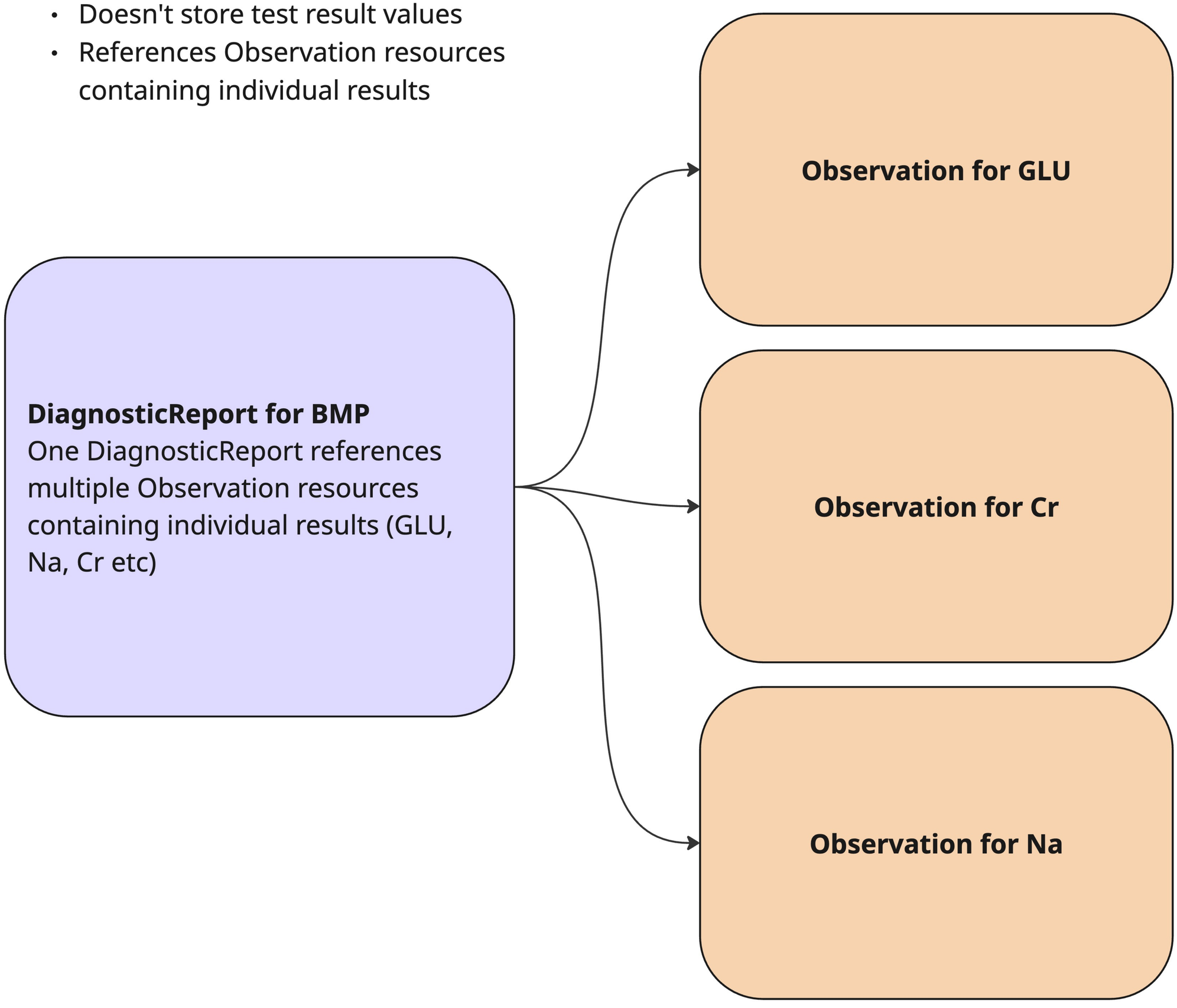
FHIR Resources

DiagnosticReport

- One per test panel
- Groups Observation resources and provides clinical context
- Doesn't store test result values
- References Observation resources containing individual results

Observation Resource

- Contains individual test result



Note: Chronologically, Observations are created first, then grouped under a DiagnosticReport via references.

Data pipeline: raw ingest - parsed relations - validated and normalized FHIR artifacts

Immutable: source of truth

Raw Data

ingestion_id (UUID, PK + FK)
content_bytes (BYTEA)
content_mime (default text/csv)
content_size_bytes

One-to-one both required

Ingestion

Data parsed from CSV into data relations

One-to-many

ingestion_id (UUID, PK)

instrument_id TEXT NOT NULL
run_id TEXT NOT NULL
Unique(instrument_id, run_id)

uploader_id, TEXT NOT NULL
spec_version, TEXT NOT NULL

uploader_received_at, timestamp NOT NULL
api_received_at, timestamp NOT NULL

submitted_sha256 TEXT nullable=True
server_sha256 TEXT NOT NULL

status, TEXT NOT NULL
error_code TEXT nullable=True
error_detail JSONB nullable=True

source_filename, TEXT NOT NULL
idempotency_disposition enum - fixed options (CREATED / DUPLICATE_IDENTICAL / CONFLICT) nullable=True

Parsed/staging representation of data. Structured but not normalized.

Panel

panel_id (UUID, PK)
ingestion_id (UUID, FK to ingestion.ingestion_id NOT NULL)
patient_id TEXT NOT NULL
panel_code TEXT NOT NULL
sample_id TEXT
collection_timestamp (timestamp NOT NULL)
UNIQUE (ingestion_id, panel_code, sample_id) - added sample_id to cover reruns

Data validated and normalized

one-to-one(0)

One-to-many

Test

test_id (UUID, PK)
panel_id (UUID, FK to panel.panel_id, NOT NULL)
row_number (INT, NOT NULL) - for tracing back to CSV
test_code TEXT NOT NULL
test_name TEXT nullable=True
analyte_type TEXT nullable=True

result_raw TEXT NOT NULL
units_raw (TEXT, NULL)
result_value_num (NUMERIC, NULL)
result_comparator (enum: "<", ">", "=")

ref_low_raw (TEXT, NULL)
ref_high_raw (TEXT, NULL)
flag (TEXT, NULL)

UNIQUE (panel_id, test_code, analyte_type) - covers duplicates, diluted, calculated etc for samples

one-to-one(0)

Normalized clinical artifacts. FHIR-shaped persistence

Normalized Data

diagnostic_report

diagnostic_report_id UUID, PK
ingestion_id (UUID, FK to `ingestion.ingestion_id`, NOT NULL)
panel_id (UUID, FK to `panel.panel_id`, NOT NULL, UNIQUE)
patient_id TEXT NOT NULL
panel_code TEXT NOT NULL
effective_at (timestamp NOT NULL) (same as panel.collection_timestamp)
issued_at (timestamp NOT NULL) (time of normalization)
resource_json JSONB nullable=True
status enum:'final' nullable=True

Index:
diagnostic_report.patient_id

observation

Identity/provenance:
observation_id (UUID, PK)
test_id (UUID, FK to `test.test_id`, UNIQUE, NOT NULL)
diagnostic_report_id (UUID, FK to `diagnostic_report.diagnostic_report_id`, NOT NULL)
ingestion_id (UUID, FK to `ingestion.ingestion_id`, NOT NULL)
patient_id TEXT NOT NULL

Clinical content:
code TEXT NOT NULL
display TEXT nullable=True
effective_at datetime nullable=True
value_num NUMERIC nullable=True
value_text TEXT nullable=True
comparator ENUM (use result_comparator_enum) nullable=True
unit (TEXT nullable=True)
ref_low_num (NUMERIC nullable=True)
ref_high_num (NUMERIC nullable=True)
flag (TEXT nullable=True)
status (TEXT nullable=True)
resource_json (JSONB nullable=True)
UNIQUE (diagnostic_report_id, code)

Indexes:
index observation.patient_id
index observation.diagnostic_report_id
index observation.code

One-to-many(0)

Append-only operational history

Audit/processing Log
append only.
ordered by time,
keys by ingestion_id

processing_event

Event identity
event_id (UUID, PK)
ingestion_id (UUID, FK, NOT NULL)
sequence BIGINT NOT NULL (define as global sequence), UNIQUE
dedupe_key TEXT nullable = True

What the event is about
target_type (ENUM: ingestion | panel | test | diagnostic_report | observation | ai_annotation), nullable = True
target_id (UUID) points to the id of specific row in target table, nullable = True
Note: If target_type == "ingestion", then target_id == NULL (as we already have ingestion_id in this row)

WHEN
event_type NOT NULL, ENUM: see * table for enums
occurred_at NOT NULL timestamp

WHO
actor (service name) - (NOT NULL, enum: ingestion-api, parser, validator, normalizer, ai-worker)

actor_version (TEXT, nullable = True). service version / git SHA

artifact_versions JOSNB (normalizer version, schema version, prompt/model version), nullable = True

Explanation
severity ENUM: INFO, WARN, ERROR, nullable = True
message (TEXT) - human readable, nullable = True
details (JSONB), nullable = True

Constraints
UNIQUE (ingestion_id, event_type, dedupe_key) - if dedupe_key present

Indexes
(ingestion_id, sequence)
(ingestion_id, event_type)
(target_type, target_id)

AI augmentation artifacts

Stores embedded vectors for reference ranges, clinical interpretation guidelines, historical normalized lab results

AI annotation

Identifiers
ai_annotation_id UUID PK
ingestion_id UUID FK NOT NULL
target_type (ENUM: DIAGNOSTIC_REPORT, OBSERVATION, nullable = True)
target_id (UUID: points to the chosen table's PK, nullable = True)

Annotation
annotation_type ENUM ("anomaly_flag", "possible_interference", "followup_suggestion"), nullable = True

content_json (schema enforced output) JSONB NOT NULL

Traceability
provider (of LLM, eg Bedrock) (TEXT nullable = True)
model_id TEXT nullable = True
prompt_version TEXT nullable = True
temperature TEXT nullable = True
content_schema_version TEXT, NOT NULL
input_hash TEXT nullable = True
created_at timestamp nullable = True

Status
validation_status ENUM: PENDING, ACCEPTED, REJECTED. nullable = True
validated_at timestamp nullable = True
rejection_reason TEXT nullable = True

Indexes
(target_type, target_id)
(annotation_type)
(validation_status)

Vector Store (embeddings + metad

embedding_id (UUID, PK)
embedding VECTOR NOT NULL

Source
source_type NOT NULL (enum:DOCUMENT, OBSERVATION, DIAGNOSTIC_REPORT
)
source_id (UUID/TEXT NOT NULL: the PK of the corresponding table)
chunk_index INT NOT NULL DEFAULT 0 (derived from document.content)
chink_text TEXT NOT NULL (derived from document.content)
content_hash TEXT NOT NULL

Traceability
embedding_model TEXT NOT NULL
pipeline_version TEXT NOT NULL
created_at timestamp TIMESTAMP NOT NULL

Constraints
UNIQUE (source_type, source_id, chunk_index, embedding_model, pipeline_version)

Documents for vector embeddings

document
doc_id (UUID, PK)
title (Text, NOT NULL)
target_type ENUM/TEXT NOT NULL: PANEL or ANALYTE (NOT FK!!)
target_code (Text, NOT NULL) Note:(if PANEL, panel_code, if ANALYTE, analyte_code - it's NOT FK!!)
content TEXT, NOT NULL
content_format (TEXT, default text/plain)
created_at TIMESTAMP NOT NULL
last_updated_at TIMESTAMP NOT NULL
content_hash TEXT, NOT NULL

*Event types for processing_event table:
Ingestion
INGESTION_ACCEPTED
INGESTION_DEDUPED_IDENTICAL
INGESTION_CONFLICT
Parse
PARSE_STARTED
PARSE_SUCCEEDED (include row counts)
PARSE_FAILED (include error summary)
Validation
VALIDATION_STARTED
VALIDATION_SUCCEEDED
VALIDATION_FAILED (include counts + top error codes)
Normalization
NORMALIZATION_STARTED
NORMALIZATION_SUCCEEDED
NORMALIZATION_FAILED
AI augmentation
AI_ENRICHMENT_STARTED
AI_ENRICHMENT_SKIPPED (reason)
AI_ENRICHMENT_SUCCEEDED
AI_ENRICHMENT_FAILED